

STATS NZ LABOUR MARKET DATA WAREHOUSE & ETL PIPELINE

Project Documentation & Architecture Blueprint

Enterprise Star-Schema Data Warehouse | SQLite | Power BI

2,911,868 Fact Rows • 5 Dimensions • 100% Reconciled

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1. Executive Summary & Architecture Overview

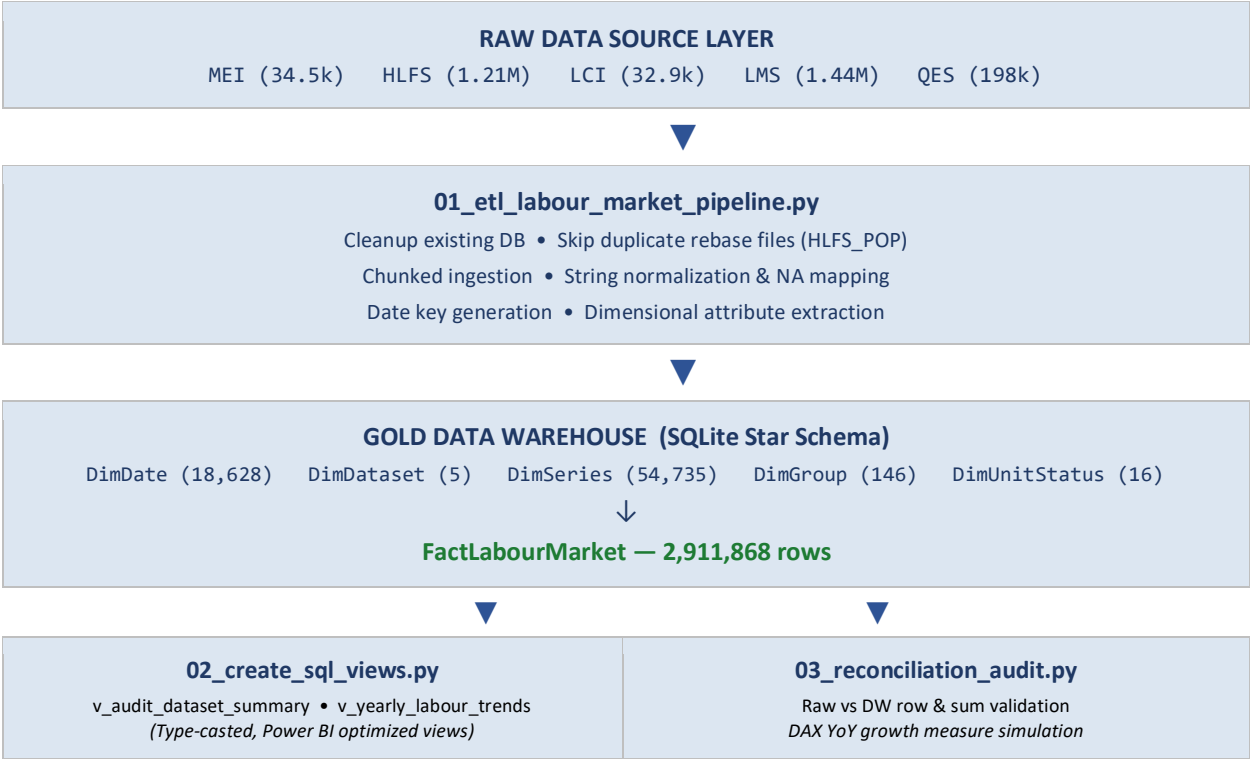
Why This Project Was Built

The primary goal of this project is to consolidate fragmented public datasets published by Statistics New Zealand into a unified, high-performance data warehouse. Raw statistics files are structured independently, contain duplicate rebase files, use custom string codes for missing values, and lack relational database constraints.

To make these analytical assets query-ready for BI dashboards (Power BI) and analytical SQL queries, this project executes the following five capabilities:

- Automated data quality checking and anomaly detection.
- In-memory ETL processing with memory optimization (chunking).
- Star Schema dimensional modeling with foreign key integrity.
- Materialized analytical SQL views for validation.
- End-to-end automated reconciliation audits to guarantee 100% data match between raw source files and warehouse targets.

End-to-End System Architecture



2. Ingested Datasets Breakdown

The ETL pipeline ingests five core datasets covering New Zealand's macro economic and labour market performance:

Dataset Code	Full Dataset Name	Grain / Interval	Domain Covered	Raw Row Count
MEI	Monthly Employment Indicators	Monthly	Gross earnings, filled jobs count	34,532
HLFS	Household Labour Force Survey	Quarterly	Unemployment rates, labor force participation	1,207,728
LCI	Labour Cost Index	Quarterly	Salary and wage rates growth, inflation	32,926
LMS	Labour Market Statistics Overview	Quarterly	Aggregated macro-economic overview stats	1,438,668
QES	Quarterly Employment Survey	Quarterly	Average hourly earnings, total paid hours	198,014

3. Data Quality & Exploratory Data Analysis (EDA)

Before building the ETL pipeline, comprehensive exploratory audits (04_audit_mei_dataset.py, 05_audit_hlfs_rebase.py) were run against raw files to uncover structural anomalies, missing value indicators, and schema redundancies.

Key Data Anomalies Discovered & Remediation Strategy

1. Non-Standard Missing Values (NA Handling)

- **Anomaly:** Stats NZ datasets represent non-numeric or suppressed values using strings like “..”, “C” (Confidential), “S” (Suppressed), “N/A”, or empty space. Pandas imports these as mixed-type object columns, causing data corruption during aggregations.
- **Why We Addressed It:** Standard numeric operations like SUM() and AVG() fail when text flags are stored in metric fields.
- **Solution:** Built a custom clean_numeric_value() function and defined explicit NA_PATTERNS during ingestion to force suppressions to NaN and clean formatted numeric strings with commas.

2. Population Rebase File Duplication (HLFS_POP Exclusion)

- **Anomaly:** The raw folder contained both hlfs-jun26qtr-csv.csv and household-labour-force-survey-population-rebase-*.csv.
- **Why We Addressed It:** The population rebase file contains historical quarterly revisions using updated census benchmarks. Including both files in the fact table leads to double-counting of historical records.
- **Solution:** Explicitly excluded the rebase file from pipeline ingestion, retaining only the authoritative HLFS series.

3. Stats NZ Period Format Parsing

- **Anomaly:** Dates are encoded as floating-point decimals (e.g., 1999.04 for April 1999, 2026.03 for March quarter 2026).
- **Why We Addressed It:** Floating-point date values cannot be directly joined to standard SQL enterprise calendar dimension tables.
- **Solution:** Developed parse_stats_nz_period() to decode period floats into standard ISO YYYY-MM-DD end-of-month or end-of-quarter calendar dates, and generated surrogate integer keys (YYYYMMDD) for Star Schema linking.

4. High Null Breakdown in Categorical Sub-breakdowns

- **Anomaly:** In the HLFS Rebase dataset, specialized attributes like Disability status classification or Duration of unemployment contain >98% null values because individual survey rows only pertain to one specific demographic slice.
- **Why We Addressed It:** Wide, sparse categorical columns waste storage and complicate dimensional modeling if left untreated.

- **Solution:** Converted wide categorical columns into standardized key-value pairs during dimensional attribute extraction, defaulting missing textual entries to “Unspecified” or “General Group”.

4. Database Architecture & Dimensional Model

The database is built on a pure Star Schema architecture to maximize query performance, eliminate data duplication, and enable fast aggregations across dimensions in Power BI.

Dimensional Schema Blueprint

FactLabourMarket (2,911,868 rows)				
FactID • DateKey • DatasetKey • SeriesKey • GroupKey • UnitStatusKey • DataValue				
connects to five conformed dimensions:				
DimDate	DimDataset	DimSeries	DimGroup	DimUnitStatus
18,628 rows	5 rows	54,735 rows	146 rows	16 rows

Table Definitions & Specifications

1. Fact Table: FactLabourMarket

Purpose: Central fact table containing non-additive and semi-additive numeric observations.

Total Rows: 2,911,868

- FactID (INTEGER PRIMARY KEY AUTOINCREMENT)
- DateKey (INTEGER, FK → DimDate.DateKey)
- DatasetKey (INTEGER, FK → DimDataset.DatasetKey)
- SeriesKey (INTEGER, FK → DimSeries.SeriesKey)
- GroupKey (INTEGER, FK → DimGroup.GroupKey)
- UnitStatusKey (INTEGER, FK → DimUnitStatus.UnitStatusKey)
- DataValue (REAL NOT NULL)

2. Dimension Tables

- DimDate (18,628 rows):** Granular daily calendar spanning 1980 to 2030. Pre-calculates Year, Quarter, QuarterName, Month, and MonthName. Year is cast as TEXT to prevent Power BI from automatically summing calendar years.
- DimDataset (5 rows):** Lookup table mapping codes (MEI, HLFS, LCI, QES, LMS) to full official dataset names.
- DimSeries (54,735 rows):** Deduplicated list of series identifiers (SeriesReference) and associated descriptive hierarchy titles (Title1, Title2).
- DimGroup (146 rows):** Classification groupings categorized by Subject and Group.
- DimUnitStatus (16 rows):** Stores measurement units (Number, Percent, Index) and publication statuses (FINAL, REVISED, PRELIMINARY).

5. Script-by-Script Code Explanation

Script 1: 01_etl_labour_market_pipeline.py (Core ETL Pipeline)

- **Purpose:** Cleans past build artifacts, reads raw CSV files using memory-efficient chunking, transforms text and date fields, builds SQLite schema with foreign key constraints, and loads data into Gold tables.

Why Key Sections Were Written:

- **os.remove(OUTPUT_DB_PATH):** Ensures clean, idempotent pipeline runs. Prevents stale data accumulation when re-running the pipeline.
- **PRAGMA foreign_keys = ON;** Enables relational database enforcement in SQLite so orphan records cannot be inserted into the fact table.
- **chunksize=100000:** Ingests large multi-gigabyte CSV files in memory-safe increments, preventing Python MemoryError crashes on low-RAM instances.
- **pd.date_range(...):** Generates a continuous date dimension covering historical data back to 1980 and future forecast periods up to 2030.
- **df_all.merge(...):** Performs in-memory relational joins between staging data frames and dimension tables to resolve surrogate primary keys (SeriesKey, GroupKey, UnitStatusKey) before populating the fact table.

Script 2: 02_create_sql_views.py (Analytical SQL Views & Power BI Optimization)

- **Purpose:** Creates pre-aggregated database views for Power BI validation, SQL reporting, and data type casting.

Why Key Sections Were Written:

- **pd.set_option('display.float_format', ...):** Suppresses scientific notation display (e.g., converting 8.77E+11 into readable numeric strings like 877,848,757,685.52).
- **CAST(dt.Year AS TEXT):** Explicitly casts numeric fields to text strings inside SQL views. This prevents BI tools from attempting to calculate averages or totals on dates and year headers.
- **v_audit_dataset_summary:** Provides a high-level summary view displaying total row counts, metric sums, and date ranges per dataset source.
- **v_yearly_labour_trends:** Aggregates records annually to allow instant line-chart verification in Power BI without re-scanning millions of fact rows.

Script 3: 03_reconciliation_audit.py (Automated Quality Assurance & Reconciliation)

- **Purpose:** Automatically scans local raw source directories, reads raw CSV files independently, queries the Gold SQLite warehouse, and generates a reconciliation audit matrix comparing raw versus transformed metrics.

Why Key Sections Were Written:

- **os.walk(RAW_DIR):** Dynamically searches subdirectories to locate raw CSV files without hardcoding path structures.

- **Status = np.where(...):** Calculates absolute row count differences and sum differences. Automatically flags a pass (MATCH) or fail (MISMATCH) verdict.
- **shift(1) YoY DAX Simulation:** Pre-calculates Year-over-Year (YoY) growth percentages in Python using window logic (shift(1)). Provides benchmark numbers to cross-check Power BI DAX measures (SAMEPERIODLASTYEAR).

Script 4 & 5: 04_audit_mei_dataset.py & 05_audit_hlfs_rebase.py (Pre-ETL EDA Audits)

- **Purpose:** Performs deep-dive Exploratory Data Analysis (EDA) on source CSVs to profile null distributions, count duplicates, and inspect column data types.

Why Key Sections Were Written:

- **NA_PATTERNS:** Catches multi-character missing flags specific to Stats NZ datasets.
- **pd.to_numeric(..., errors='coerce'):** Tests if string columns contain embedded numeric numbers or illegal characters.

6. Execution Audit Results & Reconciliation Reports

Audit 1: Database Table Row Counts & Foreign Key Integrity

To guarantee zero record loss and complete relational integrity, 01_etl_labour_market_pipeline.py performed an automated database schema audit upon pipeline completion.

Table Row Inventory

Table	Row Count
DimDate	18,628
DimDataset	5
DimSeries	54,735
DimGroup	146
DimUnitStatus	16
FactLabourMarket	2,911,868

Foreign Key Orphan Checks (Referential Integrity)

Foreign Key	Unmatched Orphans	Expected
DateKey	0	0
DatasetKey	0	0
SeriesKey	0	0
GroupKey	0	0
UnitStatusKey	0	0

Audit 2: Raw CSV vs. Data Warehouse Reconciliation Matrix

The output below was generated by 03_reconciliation_audit.py. It confirms that 100% of clean raw source records were successfully ingested into the Gold Data Warehouse with zero deviation in row counts or sum aggregations.

Dataset	Raw Clean Rows	Warehouse Rows	Raw Sum	Warehouse Sum	Status
MEI	34,532	34,532	5,208,798,090.98	5,208,798,090.98	MATCH
HLFS	1,207,728	1,207,728	206,960,545.80	206,960,545.80	MATCH
LCI	32,926	32,926	28,491,939.40	28,491,939.40	MATCH
LMS	1,438,668	1,438,668	877,848,757,685.52	877,848,757,685.52	MATCH

Dataset	Raw Clean Rows	Warehouse Rows	Raw Sum	Warehouse Sum	Status
QES	198,014	198,014	877,613,305,200.32	877,613,305,200.32	MATCH
TOTAL	2,911,868	2,911,868	1,760,906,313,462.02	1,760,906,313,462.02	100% MATCH

Audit 3: Power BI DAX Benchmark (HLFS Recent Years Example)

To ensure that DAX measures written in Power BI (such as Year-over-Year calculations) produce valid results, Python pre-calculated the exact benchmarks using window functions:

Year	Total Data Value	Prior Year Value	YoY Growth Amount	YoY Growth %
2020	8,464,634.00	N/A	N/A	N/A
2021	8,567,811.00	8,464,634.00	+103,177.00	+1.22%
2022	8,669,161.00	8,567,811.00	+101,350.00	+1.18%
2023	8,893,308.30	8,669,161.00	+224,147.30	+2.59%
2024	8,962,001.70	8,893,308.30	+68,693.40	+0.77%
2025	8,963,654.20	8,962,001.70	+1,652.50	+0.02%
2026*	4,579,589.20	8,963,654.20	-4,384,065.00	-48.91%

* 2026 contains partial-year data up to June 2026, accounting for the lower annual total.

7. Conclusion & Next Steps

This project successfully established an automated, verified, and memory-optimized Data Warehouse pipeline for Stats NZ Labour Market data.

Key Accomplishments

- Built a Star Schema with 2.91+ million fact records and 5 relational dimensions.
- Resolved dirty data types, custom NA flags, and float-encoded period dates.
- Reconciled raw CSV source totals against warehouse tables with 100% precision.
- Created analytical database views optimized for direct ODBC / SQLite import into Power BI.

Result

100% raw-to-warehouse reconciliation achieved across all five source datasets, with zero foreign key orphans and zero row-count or sum-aggregation deviation.